nature portfolio

Peer Review File

Clinical Utility of Receptor Status Prediction in Breast Cancer and Misdiagnosis Identification Using Deep Learning on Hematoxylin and Eosin-Stained Slides

Corresponding Author: Dr Gil Shamai

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

“3.2. Lines 702-713: I don't see the code on github for that part.
Thank you for pointing this out. We did not include the code for the proposed heatmap method on GitHub, as we did not consider it a novel contribution of our study. As mentioned in line 713 of the original manuscript, methods like GradCam yielded similar outcomes. However, if the reviewer feels it is necessary, we are happy to provide the code upon acceptance of the manuscript.”

--> Yes, it would be best to have everything in the github repository to facilitate reproduction by the users of the full study.

“Train and test": as such, it seems like we could fear data leaks for the study related to these datasets, right?
We appreciate your concern regarding potential data leaks. Carmel-Test-Rescanned consists of slides from the Carmel-Test dataset that were rescanned using a different scanner. However, your concern regarding Carmel-Intratumor is valid. Carmel-Intratumor was designed to include patients from the entire Carmel dataset, not just Carmel-Test, to broaden the analysis. Here are some clarifications: (1) While patients from Carmel-Train are included in Carmel-intratumor, the system was applied to H&E slides from different blocks than those used in Carmel-Train. (2) Carmel-Train consists of five-folds, and we ensured that each model was applied to patients it was not trained on. While there is a potential risk of overfitting during hyperparameter tuning, this tuning was conducted solely using the TCGA training folds, not Carmel data. (3) The risk of overfitting in this context does not lead to an advantage. The system was applied to the Carmel-Intratumor dataset to identify tumor blocks predicted as ERpositive and PR-positive, even when the ground truth was negative. Overfitting to the ground truth status would actually result in identifying fewer potential false negative cases, which is contrary to our objectives.”

--> Were this clarified in the manuscript as well? If not please add it.

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here are my thoughts on this response to reviewer 2
- First, the AI performance is compared to pathologists' performances looking visually at "tile" images. This should also be done on "slide" images, since pathologist usually do these predictions looking at the whole slide, not single tile. It makes the comparison unfair.
**** Answer to Question 2:
- Second, the authors say "Although this comparison provides interesting insights, it is not the main focus of our study, and we did not add it to the paper."
"I would also encourage the authors to perform a blinded study..."
Since reviewer 2 said Things like that would definitely increase the novelty of the study", I would suggest to add it to the

manuscript (providing they add a comparison with estimation from WSI in addition to tiles)
**** Question 4:
"From a clinical perspective, the three biomarkers are typically used in combination to categorize patients ...."
Question is addressed in the rebuttal latter but the manuscript does not seem to have been altered accordingly.
*** Question 5:
"-The basis of the predictions—whether they pe.... "
same as before - the authors list 5 points to justify they choice - this should be added more clearly as such in the manuscript's method section as well

Reviewer #2

(Remarks to the Author)
thank you for addressing all of my comments

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
Thanks for addressing the comments and adding the details to the papers. I have no more comment.

To all reviewers

Please find below a point-by-point response to all of your comments referring to the relevant changes made in the manuscript. We also attach a copy of the revised manuscript with all modifications in color for your convenience.

Reviewer #1 (Remarks to the Authors, Answers)

In this study, the authors propose a method to predict hormone receptor status and HER2 status from hematoxylin and eosin (H&E) stained pathology slides using deep learning. This is a common problem in computational pathology which was already published six years ago! However, the introduction lacks references to prior work. It gives the impression that the authors are the first to address this issue, which is not the case. Previous studies have demonstrated high performance in predicting these biomarkers from H&E stained slides using deep learning, and products exist in the market for this purpose. Consequently, the introduction is strongly misleading and requires complete rephrasing. For your reference, these are some of the studies on exactly the same task, dating back to 2018, yielding a high performance: Couture et al PMID 30182055, Kather et al PMID 33763651, Fu et al PMID 35122049, Arslan et al biorxiv doi 10.1101/2022.01.21.477189. Planned or available products by Paige (https://paige.ai/paige-answers-call-to-better-identify-breast-cancer-patients-with-low-expression-of-her2/) and Panakeia (https://www.panakeia.ai/products, PanProfiler Breast)

Thank you for your feedback. We acknowledge that many studies in recent years have demonstrated that ER, PR, and ERBB2 statuses can be predicted using deep learning analysis of H&E images. In fact, our own study (Shamai et al., 2019) was among the first to explore this. However, existing studies have primarily focused on overall prediction performance (often via AUC) rather than the practical utility of such systems in clinical practice. Our study aims to fill this gap by exploring the potential impact and utility of predicting ER, PR, and ERBB2 statuses in a clinical setting.

Following your feedback, we conducted a comprehensive search on Google Scholar for papers published since 2018, using specific search terms related to receptor prediction from H&E images in breast cancer. This search yielded 1,620 articles, from which we identified 31 peer-reviewed papers that addressed the prediction of receptor expression from (H&E) images in breast cancer. Analysis of these 31 papers showed that over the past six years, there has been a growing interest in adopting machine learning models for predicting receptor expression from H&E images in breast cancer. Nevertheless, the overall performance of machine learning models remains inferior to the standard of care molecular profiling methods using IHC. Therefore, despite the potential of H&E-based receptor prediction to improve breast cancer diagnosis, it remains unclear how machine learning models can be integrated into clinical workflows and their actual impact on diagnosis. This highlights the need for further applicative evidence coupled with robust clinical validation, which would require extensive real-world datasets from diverse populations and multiple sites, as well as access to patients' tumor blocks for re-diagnosis. Access to such data was perhaps the largest barrier in previous studies, which were limited by small sample sizes and reliance on a few publicly available datasets.

It is important to emphasize that our primary contribution **is not** to improve receptor prediction performance or develop a new deep-learning methodology. Rather, our study presents a clinical validation study for the potential impact and utility of ER, PR, and ERBB2 status prediction from H&E images in clinical practice, by introducing three practical applications:

1. **Hormone Therapy Eligibility:** We show that for about a third of breast cancer patients, hormone therapy eligibility can be determined from H&E, non-inferior to IHC-based interpretation.
2. **Detecting False Negatives:** We deploy an H&E-based analysis system to detect false negative diagnoses in hormone status by restaining and reassessing tumor samples flagged as potentially misdiagnosed.
3. **Evaluating Unstained Tumor Regions:** We evaluate unstained tumor regions for detecting false negative diagnoses in ER and PR due to intra-tumor heterogeneity.

Our analysis highlights substantial variabilities in IHC-based diagnoses and suggests that the adoption of H&E-based models could lead to the detection of thousands of false negative receptor status diagnoses in breast cancer every year. Given the current molecular profiling guidelines' strong emphasis on reducing false negative diagnoses, our findings could impact future decisions about integrating AI-based H&E analysis into clinical workflows.

In our introduction, we cannot cite all 31 papers on receptor prediction in breast cancer. Instead, we cited what we considered the most relevant and impactful studies. The citations you suggested focus on overall receptor prediction performance. Some of them do not address ER, PR, and ERBB2 prediction in breast cancer specifically, and those that do often report lower performance (e.g., around 80-85% for ER). Including citations for studies with lower performance might not add significant value. Additionally, the presence of commercial products does not diminish the need for research into the clinical impact of these applications.

Following your comment, we added a few sentences in the introduction to clarify the contribution (lines 61-65 in the revised manuscript).

The study is potentially interesting due to its large training set of 19,000 labeled H&E slides. However, in computational pathology, the focus is typically on the number of patients rather than the number of slides. Usually, a single representative slide per patient suffices, and adding more slides per patient does not significantly improve performance -- this is also what the authors find in this paper, confirming previous work (which they fail to cite). The authors overall include data from 7,950 patients, which exceeds many studies, but there are others with larger patient cohorts in computational pathology research (e.g. Campanella et al Nat Med 2019, or Wagner et al Cancer Cell. 2023).

Thank you for your feedback. While the papers mentioned do have larger datasets, they address different problems in digital pathology. For instance, Campanella et al. focused on predicting cancer presence rather than receptor status, and Wagner et al.'s study did not involve breast cancer.

We agree that the number of patients is important, as each patient serves as an independent observation. However, we would like to emphasize a few points:

1) Even when considering the number of patients, our study represents the largest cohort for receptor prediction in breast cancer.
2) Not only is the number of patients important, but also the diversity of datasets, inclusion of external cohorts, and the use of recent, unbiased real-world data. Particularly important is the inclusion of data that enables re-diagnosis of cases.
3) It is also important to acknowledge the number of IHC diagnoses, which exceeds the number of patients due to the presence of multiple tumors per patient, either at the time of diagnosis or over different periods.
4) The large number of slides allowed us to conduct additional analyses, such as evaluating whether multiple slides per block enhance performance.

Lines 87-93 describe the authors' method, suggesting they have devised a novel workflow. However, they recapitulate standard workflows in the field, already published multiple times (e.g. in Wagner et al Cancer Cell 2023). Specifically, the combination of convolutional neural networks (CNNs) with transformers is not new. Citing the "attention is all you need" paper in reference 20 as foundational for their transformer-based multiple instance learning (MIL) is misleading. This paper is a general introduction to transformer architecture, while many many subsequent studies have applied transformers to pathology and are more relevant to the problem at hand.

We acknowledge that the combination of transformers with CNNs is not novel, and we did not intend to claim originality in this workflow. Following your suggestion, we have added more specific references that better align with our methods (lines 93-94 in the revised manuscript).

In the results section, there is inconsistent and selective reporting of statistics. For example, in line 146, the authors report only the sensitivity, but not the specificity, for a selected cohort. This cherry-picking of data limits the meaningful insights derived from the study.

We recognize the importance of providing a comprehensive view of the system’s performance. Our primary objective in this section is to demonstrate the potential of deep learning for accurately identifying HR+ patients without IHC. To this end, we focused on evaluating the positive predictive value (PPV) and specificity for HR+ identification. These measures were chosen because they directly reflect the system's practical utility.

Sensitivity, which measures the percentage of actual HR-positive slides correctly classified as HR-positive, is indeed a valuable complementary measure. However, in the context of our study, we found that the percentage of impacted patients is a better complementary measure because it serves as a more practical indicator of the system's effectiveness. Moreover, in scenarios of accurate positive classification (i.e., PPV = 1 and specificity = 1), sensitivity can be derived from the formula:

$\%Impacted\ slides \times \frac{\#Total\ slides}{\#Positive\ slides}$,

where $\%Impacted\ slides$ represents the percentage of slides above the threshold out of all slides. Thus, under conditions of accurate positive classification, sensitivity closely aligns with the %impacted patients plots, typically appearing slightly higher, and does not provide additional unique insights.

That said, we agree that including sensitivity values offers a clearer picture of the model's performance, and we have added the sensitivity values for each threshold to the plots in Figure 2.

A minor point concerns the introduction, where specific details about the used cohort are mentioned, particularly in lines 77 to 80. This information seems out of place and more suited for another section of the paper.

We appreciate your insight regarding the introduction. The inclusion of specific cohort details in the introduction was inspired by the approach taken in *Campanella et al., "Clinical-grade computational pathology using weakly supervised deep learning on whole slide images," Nature Medicine*. Since the construction of the dataset is a key aspect of our study's contributions, we believe that presenting this information in the introduction enables a clearer understanding of the paper's message and contributions.

Reviewer #2 (Remarks to the Authors, Answers)

--Numerous studies with similar methodologies have been published to date. While the authors' extensive efforts are commendable, the novelty of achieving high AUCs, particularly above 0.90 for ER status, has been previously documented. This raises questions about the novelty of their work.

Thank you for your feedback. Many studies in recent years have demonstrated that ER, PR, and ERBB2 statuses can be predicted using deep learning analysis of H&E images. However, these studies have primarily focused on overall prediction performance, often measured by AUC, rather than the clinical utility of such systems. Given that the overall performance of these models still falls short of IHC-based diagnosis, it remains unclear how H&E image analysis can impact clinical practice.

Our main contribution is not to enhance receptor prediction performance or introduce new deep-learning technologies. Instead, we aim to evaluate and validate whether and how deep learning-based predictions of ER, PR, and ERBB2 statuses from H&E images could impact clinical practice. To this end, we explore three potential applications where this methodology could enhance diagnosis:

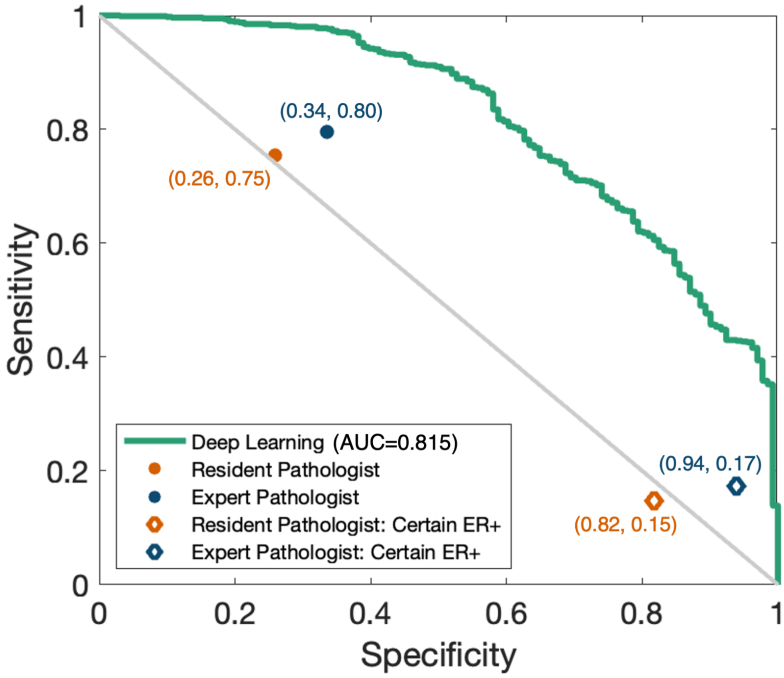
1. Despite the overall performance being lower than IHC-based diagnosis, we show that for about a third of the patients, it is possible to accurately determine hormone therapy eligibility using H&E images alone.

2. We show that H&E-based diagnosis can be used as a second-read system for detecting potential false negative diagnoses.
3. We show the utility of H&E-based models when applied to the entire tumor to further enhance diagnosis.

Given the current emphasis in the guidelines on mitigating false negative diagnoses in molecular profiling, the second and third applications are especially important, as they may impact future decisions regarding the integration of AI-based H&E analysis in clinical workflows. Following your comment, we have added a few sentences to the introduction to clarify our contributions relative to other studies (lines 61-65 in the revised manuscript).

I would also encourage the authors to perform a blinded study where the machine competes with the pathologist in determining ER status based on an H/E image. Pathologist usually are quite good at estimating if a tumor will be ER positive by IHC. The question then is how much better the machine is compared to a well-trained pathologist. Things like that would definitely increase the novelty of the study.

Thank you for your suggestion. We conducted an additional study to compare the performance of pathologists with our deep learning model in predicting ER status from H&E images. We randomly selected 943 H&E tile images from the Carmel-Test dataset and asked both a resident pathologist and an expert pathologist to predict the ER status from these images. To specifically assess their ability to identify ER-positive cases (in line with our first proposed application), we also asked them to mark only the H&E images they were most certain were ER-positive. We then compared their results to the performance of our deep learning model.



Our findings indicate that, while pathologists can infer patterns in the H&E images correlated with ER status, their performance does not match that of the deep learning model. Specifically, no features detected by the resident or expert pathologist could guarantee ER positivity. In contrast, the deep learning model was able to identify images that are almost certainly ER-positive with high specificity, even within grade 3 tumors.

It is important to note that, even if some pathologists could enhance their skills to perform this task well, predicting receptor status from H&E images by pathologists is unlikely to be standardized or recommended in clinical guidelines. In contrast, a computerized tool can be standardized and validated for consistent and reliable use.

Although this comparison provides interesting insights, it is not the main focus of our study, and we did not add it to the paper. If the reviewer wishes, we would be glad to include this analysis in the paper.

--While the predictions for ER status are impressive, the results for PR and HER2 status fall short (i.e., AUCs in the external validation datasets are below 0.83). This limitation is not adequately acknowledged in the study. Given the significant therapeutic implications of HER2 status, this represents a notable drawback.

Indeed, the ability of deep learning models to predict PR and HER2 status from H&E images is generally lower than their ability to predict ER status. This trend is observed across various studies in the field. We believe this is an inherent limitation of the biology rather than a limitation of the models. We agree that this limitation should be clarified and adequately acknowledged in the discussion. Following your comment, we have added this to the discussion section (lines 527-533 in the revised manuscript).

--From a clinical perspective, the three biomarkers are typically used in combination to categorize patients into HR+/HER2- disease, HER2+ disease (further subdivided into HR+ and HR-), and triple-negative breast cancer. The authors have not demonstrated whether their method can accurately classify breast cancer into these three clinically relevant groups with AUCs greater than 0.95. Absent this capability, the clinical applicability of their approach may be confined to ER status prediction, an area that lacks innovation.

Thank you for your feedback. We acknowledge the importance of categorizing patients into the four breast cancer subtypes. However, this categorization requires accurate prediction of HER2 status, which remains a limitation of H&E-based predictions. Additionally, our method does not aim to replace IHC entirely. Instead, we demonstrate that while the overall performance of deep learning may not fully substitute for IHC, it can still be valuable for identifying patients eligible for hormone therapy, and for identifying potential receptor misclassifications. Our approach thus offers a practical application of machine learning in clinical practice, despite the current limitations of accurately categorizing breast cancer based on H&E images.

--The basis of the predictions—whether they pertain to slides, blocks, or patients—remains unclear. It is crucial that predictions in the external validation datasets, which represent true validations, are made at the patient level, not the slide level. This is because clinical decisions are informed by the aggregate results of each biomarker per patient.

Thank you for your insight. We agree that clinical decisions are informed at the patient level. However, there are several reasons why we reported results per slide rather than per patient:

1. **Multiple Labels per Patient:** Some patients had multiple IHC results showing distinct receptor statuses due to intratumor heterogeneity, multiple lesions with different molecular profiles, or even multiple tumor excisions at different times. For instance, a core needle biopsy (CNB) and a subsequent surgical excision might show different receptor statuses (as seen in Fig 3a). In some cases, even tumors within the same patient had multiple IHC results.
2. **Slide-Level Prediction:** The deep learning system predicts receptor expression based on the tumor region visible in the H&E slide. This localized prediction does not necessarily reflect patient-level expression, which could be influenced by other regions of the tumor or other tumors.
3. **Clinical Application:** The system is designed to be applied per H&E slide, where each result represents the slide itself and perhaps its local environment (block). For patients with multiple tumors, pathologists benefit from seeing predictions per tumor to inform decisions. Another example is that applying the system immediately to a biopsy is preferable to waiting for additional surgical specimens for aggregation. Therefore, if the system would be applied per slide, the analysis should report the performance of the desired application per slide.
4. **Inconsistency Across Datasets:** Inconsistent labeling across datasets further complicates patient-level analysis. For example, TCGA often provides a single IHC result, while datasets like Carmel offer multiple IHC diagnoses in many cases.
5. **Block-Level Reporting:** Reporting per block instead of per slide is an alternative, but as demonstrated, aggregating slide scores from the same block provides no added benefit since they are usually similar. Additionally, some datasets lack multiple slides per block, adding to the inconsistency of performance measurement.

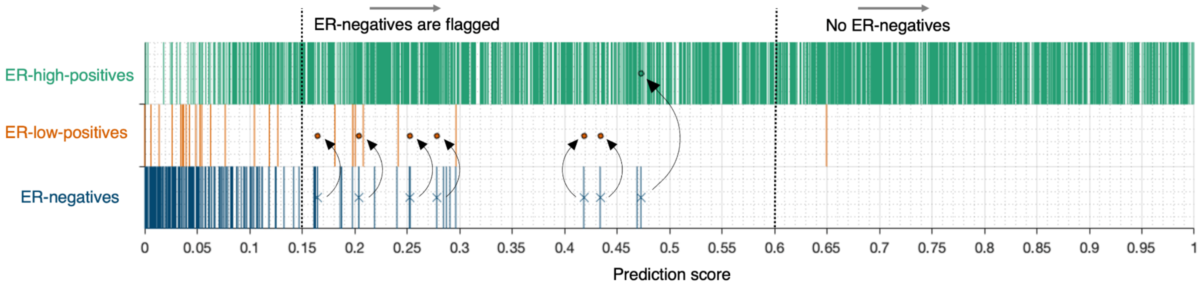
Please note that for accurate statistical analysis, confidence intervals and P values were calculated assuming independence at the patient level. For example, bootstrapping was done at the patient level to maintain statistical rigor.

For the quality assurance application tested at Carmel Medical Center, where we aimed to flag potential misclassifications, each block was stained once for IHC, representing the block's receptor status. Thus, the decision to flag a case for reclassification had to be made at the block level. For tumor blocks with more than one H&E slide, we determined if a block should be flagged for reclassification by averaging the prediction scores of its slides. Notably, we observed no differences in the flagged blocks when, instead of averaging, we randomly selected a single H&E slide per block and used its score. Following your comment, we have clarified this methodology in the Methods section (lines 746-749 in the revised manuscript).

--The authors note that ER and PR levels are quantitatively measured via IHC, a critical detail for clinicians, particularly when ER and PR levels range from 1-10%. In scenarios where ER and PR levels are low, the decision to recommend endocrine therapy remains ambiguous according to the latest ASCO guidelines. This subset of tumors warrants further investigation by the authors.

Thank you for your suggestion. The updated clinical guidelines indeed consider ER-IHC of 1-10% as ER-low positives and require reporting these cases, as the benefit of endocrine therapy is unclear for this subset. To the best of our knowledge, a PR-low-positive group was not defined or noted in the guidelines.

Following your comment, to shed more light on the ER-low positive subset, we visualized the ER status (ER-negative, ER-low-positive, ER-high-positive) of tumor blocks from the Carmel Medical Center in relation to their prediction scores, with slide scores averaged per block (Supplementary Fig 2. And lines 235-239 in the revised manuscript).



Each tumor block is represented by a line segment according to its prediction score and ER status (ER-negative, ER-low-positive, or ER-high-positive). ER-negative cases with scores above the threshold of 0.15 were flagged for reassessment. The reclassified ER-negative cases are marked with an X and an arrow pointing to their newly assigned ER status.

As observed, above a certain threshold, the system effectively identifies ER-positive cases, as demonstrated in the first proposed application in Figure 2. Additionally, the system almost exclusively identifies ER-high-positives. ER-negative cases with prediction scores above 0.15 were flagged for reevaluation, aligning with the second proposed application for quality assurance. The re-diagnosed cases were predominantly ER-low-positives, as reported in Extended Data Table 2a. This indicates that the quality assurance application is particularly adept at identifying ER-low-positives, which are more challenging to diagnose and often missed by pathologists. This also demonstrates that ER-low-positives with intermediate prediction scores were occasionally missed by IHC staining or pathologist interpretation.

--Similarly, the clinical significance of HER2 status, categorized as 0, 1+, 2+, and 3+, including HER2-low disease, should not be overlooked. The authors are encouraged to extend their focus to this category as well.

Thank you for your suggestion. Although the proposed methodology is limited for Her2 status, we agree that it would also be interesting to present the performance of the system for specifically predicting Her2-low cases:

	Carmel-Test	TCGA-Test	Haemek	CHUCB	Ipatimup
ERBB2 0 or low VS ERBB2 positive	0.844 (n=1855)	0.790 (n=183)	0.823 (n=981)	0.806 (n=176)	0.690 (n=100)
ERBB2 0 VS ERBB2 low or positive	0.622 (n=1705)	0.604 (n=118)	0.588 (n=750)	0.507 (n=176)	0.486 (n=100)
ERBB2 low VS ERBB2 0 or positive	0.519 (n=1851)	0.633 (n=81)	0.620 (n=807)	0.620 (n=133)	0.611 (n=75)

This analysis shows that it is harder to identify Her2-low based on H&E images than to identify Her2-positive cases. We have added this analysis to Supplementary Table 2.

Reviewer #3 (Remarks to the Authors, Answers)

3.1. Hyper-parameters and optimization. (Lines 614-629): only final values for the parameters are given but no information about the range explored or how they were determined is given.
We appreciate your attention to detail regarding hyperparameters and optimization. The hyperparameters were selected manually based on commonly used values and optimized according to the best cross-validation performance. The process typically involves trial and error rather than a predefined systematic search, which is why the exact range explored is not fully documented. In our study, automatic hyperparameter tuning was not conducted due to computational and time constraints, as well as the relatively minimal impact of hyperparameter variations observed in preliminary experiments. It is important to note that hyperparameters, including the number of training epochs, were determined based solely on the first two folds of TCGA-Train for ER status prediction.

3.2. Lines 702-713: I don't see the code on github for that part.
Thank you for pointing this out. We did not include the code for the proposed heatmap method on GitHub, as we did not consider it a novel contribution of our study. As mentioned in line 713 of the original manuscript, methods like GradCam yielded similar outcomes. However, if the reviewer feels it is necessary, we are happy to provide the code upon acceptance of the manuscript.

3.2. Lines 702-713: I would also appreciate some clarification (or a sup fig?) as these explanations are not very clear. For example, line 711: "processing significantly larger tiles at once" - not sure what larger tiles are if all of them are 256x256 pixels and 1 MPP.
We understand the need for clarification. The model was indeed trained on 256x256 tiles at MPP=1, and thus, it should typically be applied to tiles of the same size and MPP. However, in the heatmap calculation, we omit the average pooling operation and fully connected layer, opting instead to compute a weighted average of the tile's feature maps using the fully connected weights (following the proposed method in [13]). This approach is considered "fully convolutional", meaning that applying the calculation across all tiles in a region of SxS non-overlapping tiles is equivalent to applying it once to the entire SxS area (for instance, if S=8, this corresponds to a 4096x4096 tile at MPP=1). Following your suggestion, we have added Supplementary Fig. 3 with a caption to visualize this process.

3.3. Section line 607-611: How many tiles were generated for each cohort?
Thank you for your suggestion. We measured the number of tiles for each cohort and created histogram plots for better clarity (Supplemantary Figure 1).

3.4. Fig1: "Carmel-intratumor and Carmel-test-rescanned patients are a subset of Carmel-Train and test": as such, it seems like we could fear data leaks for the study related to these datasets, right?
We appreciate your concern regarding potential data leaks. Carmel-Test-Rescanned consists of slides from the Carmel-Test dataset that were rescanned using a different scanner. However, your concern regarding Carmel-Intratumor is valid. Carmel-Intratumor was designed to include patients from the entire Carmel dataset, not just Carmel-Test, to broaden the

analysis. Here are some clarifications: (1) While patients from Carmel-Train are included in Carmel-intratumor, the system was applied to H&E slides from different blocks than those used in Carmel-Train. (2) Carmel-Train consists of five-folds, and we ensured that each model was applied to patients it was not trained on. While there is a potential risk of overfitting during hyperparameter tuning, this tuning was conducted solely using the TCGA training folds, not Carmel data. (3) The risk of overfitting in this context does not lead to an advantage. The system was applied to the Carmel-Intratumor dataset to identify tumor blocks predicted as ER-positive and PR-positive, even when the ground truth was negative. Overfitting to the ground truth status would actually result in identifying fewer potential false negative cases, which is contrary to our objectives.

3.5. Fig 2: I'd love to see the sensitivity there too, and added to the discussion.

Thank you for your insightful comment. Following your suggestion, we have added the sensitivity values for each threshold to the plots in Figure 2. We agree that this addition provides a more comprehensive view of the model's performance. The sensitivity can be interpreted as the percentage of actual HR-positive slides that are correctly classified as HR-positive. Note that in the scenario of accurate positive classification (i.e. PPV = 1, specificity = 1), by definition, the sensitivity is equivalent to

$$\%Impacted\ slides \times \frac{\#Total\ slides}{\#Positive\ slides},$$

where $\%Impacted\ slides$ represents the percentage of slides above the threshold out of all slides. Thus, under conditions of accurate positive classification, the sensitivity closely aligns with the %impacted patients plots, typically appearing slightly higher.

Our primary objective in this section is to demonstrate the potential of deep learning for accurately identifying HR+ patients without IHC. To this end, we focused on evaluating the positive predictive value and specificity for HR+ identification, while also considering the percentage of impacted patients as a practical measure. While sensitivity is indeed important, we have chosen to limit its discussion to the figure's caption to maintain focus on the study's main goals.

It is unclear why 10 to 50% of slides were identified as positive according to Fig 2, while Fig1a shows more than 2/3 of patients should be.

Figure 1a shows that more than two-thirds of patients are identified as HR-positive based on the distribution of biomarker expression levels in the dataset, as determined by IHC. This reflects the actual prevalence of HR-positive cases in the patient cohort. In contrast, Figure 2 illustrates the system's performance in identifying HR-positive cases using varying threshold settings, which might result in a range of 10 to 50% of slides being identified as positive. Consequently, not all actual HR-positive patients are captured. If this is not the concern you were raising, I would appreciate further clarification to address your comments accurately.

3.6. Line 202: what threshold was used for. that analysis? If different from the one used in the previous section, please justify.

Thank you for pointing this out. The thresholds used for the analysis were 0.15 for ER, 0.68 for PR, and 0.91 for ERBB2. Following your comment, we have included these exact values in the Methods section (line 744 in the revised manuscript). These thresholds differ from those used in Figure 2, where a threshold of 0.6 was employed for identifying HR-positivity. The purpose of Figure 2 was to demonstrate that the system could accurately identify HR-positive cases, showing near-perfect positive classification at the chosen threshold. For the second-read system, different (typically lower) thresholds are necessary to identify cases of discordance between the system's classification and the receptor status reported in pathology reports. Please see the new Supplementary Figure 2 for a visualization of this analysis. Additionally, it is important to note that Figure 2 utilized a model that predicts HR status, whereas the second-read system used separate models for predicting ER, PR, and ERBB2 statuses. Since these are distinct models with different output score distributions, the thresholds used are consequently non-comparable.

3.7. For each dataset, it is unclear how the ground truth was obtained and how the ER/PR/ERBB2 status were assigned.
Thank you for your comment. Patient pathology data was collected from the pathology reports. ER, PR, and ERBB2 statuses were defined in accordance with current guidelines. Namely, samples with at least 1% of tumor nuclei showing positive nuclear staining by ER-IHC and PR-IHC were interpreted as ER-positive and PR-positive, respectively. For ERBB2, samples with an IHC score of 3+ or an IHC score of 2+ accompanied by a positive FISH test were considered ERBB2-positive. Following your comment, we added these details in the Methods section (lines 553-558 in the revised manuscript).

3.8. Line 291-294, Fig 4b: it is not clear how the data were aggregated per block, and if the method chosen to aggregate would have an impact on the results or not.
We appreciate your observation. As mentioned in the caption of Figure 4b, "We computed a per-block score by averaging the scores of individual H&E slides within each corresponding block." Following your comment, we have also included this detail in the main text (line 313 in the revised manuscript). In our preliminary experiments, we explored alternative aggregation methods, such as using the maximum score per block, but found that the choice of aggregation method did not impact the results.

4.1. Please report other metrics usually used for such. studies (F1-score, Precision/recall score....). Because of highly unbalanced datasets, AUC are difficult to interpret (table 1 and rest of the paper). Confusion matrices would be interesting too.

Thank you for your suggestion. While confusion matrices provide metrics such as specificity, sensitivity (recall), PPV (precision), and NPV, these metrics depend heavily on the chosen classification threshold. The classification thresholds are application-specific and can vary depending on the intended use of the system. Furthermore, without a special calibration step, the absolute value of the thresholds used by deep learning models may not have intrinsic meaning. For example, using a threshold of 0.5 to binarize the data and calculate a confusion matrix may not provide meaningful insights, and may not lead to comparable metrics between different models.

Therefore, in Figure 2, we illustrated the relationship between these metrics and the threshold, selecting 0.6 as the operating threshold based on cross-validation results for the specific application being discussed. To assess the overall performance of the system, we utilized the AUROC, which does not depend on a classification threshold or the prevalence of positive cases. However, given the imbalanced nature of the data, additional performance metrics can indeed provide a more comprehensive understanding of the model's capabilities. Following your suggestion, we calculated the AUC for both precision/recall and its complementary NPV/specificity curve. These metrics do not depend on a classification threshold, but do depend on the prevalence of positive cases, and are typically used in the case of imbalance data in addition to AUROC. The results of these additional analyses are presented in Supplementary Table 1 in the revised manuscript.

4.2. Line 297: "not statistically significant" mentioned but the test used is not mentioned.
Thank you for your comment. To assess the statistical significance of the performance gap between two classifiers applied to the same test data (as in Fig. 4b and Extended Table 3), we employed a bootstrapping method at the patient level. We then calculated the P value from the empirical distribution of the resulting performance gaps. This resulted in P = 0.182, 0.408, and 0.162 for ER, PR, and ERBB2, respectively. Following your comment, we indicated the P values (line 319 in the revised manuscript) and added this explanation to the Statistical Analysis section (Methods).

5.1. Line 285-289:"The models performed better on each target cohort when their training data included that cohort...". These observations are not obvious at all when looking at Fig 4a.

Thank you for your comment. In the ABCTB-Test subplot of Figure 4a, the green and orange bars (indicating training that included ABCTB) are higher than the blue bar (indicating training that excluded ABCTB). Similarly, in the TCGA-Test subplot, the green and orange bars are higher than the blue bar. For the Carmel-Test subplot, the blue and orange bars (indicating training that included Carmel) are generally higher than the green bar (indicating training that excluded Carmel), with one exception. To make this clearer, we have added the ABCTB example to the main text (line 305 in the revised manuscript). Additionally, the Figure's caption includes further clarifications.

5.2. Is the process performing better or worse depending on sample excision type and cancer grade?
In Extended Data Table 1, we compared the distribution of sample types (core needle biopsy [CNB] and surgical excision [EXC]) across the identified HR+ cases to the distribution of sample types across the actual HR+ cases. We observed that CNB samples had slightly higher percentages of identified HR+ cases (e.g., for Carmel-Test: 76.5% versus 72.8%). This indicates that the system is slightly more likely to identify HR+ cases within CNB samples compared to surgical excisions when adjusting for the actual HR+ percentage.

For high-grade tumors, the gap is more pronounced, showing that the system is indeed less likely to identify HR+ cases within high-grade tumors than within moderate and low-grade tumors, as mentioned in the caption. Nevertheless, a significant percentage of high-grade tumors were accurately identified as hormone-positive, despite their morphology typically not being associated with hormone-positivity. This is a notable finding. Overall, since the system could identify HR+ samples across various categories, its application should not be limited to specific breast cancer subsets.

6.1. Data exclusion reports "slides with not enough tissue were excluded". Please be more specific re what is "not enough". (also line 79 & 515 of manuscript / method)

Thank you for your feedback. We excluded slides that contained fewer than 100 tissue tiles because our transformer configuration requires 100 tiles per bag, representing 6.55 square millimeters of tissue. Such slides were rare, constituting only 0.19% of the dataset. We have added this information to the Methods section (lines 549-552 in the revised manuscript) and referenced this detail from line 79 in the original manuscript.

6.2. Line 22: "This task is among the few that humans are incapable of performing " - not sure about that.
Thank you for your insight. We revised the sentence for better accuracy (line 448 in the revised manuscript).

6.3. Abstract: please define PPV. - throughout the text, in general, please make sure abbreviations are defined the first time they're used.
Thank you for your comment. We have now replaced PPV with “Positive Predictive Value” in the abstract for clarity. Regarding other instances throughout the paper, we reviewed the text and found that all abbreviations were defined upon first use. If there are any specific instances we may have missed, we would appreciate further feedback to address them accordingly.

6.4. How long did it take to run the training on such model? How long does it take to run testing on 1 typical case on such model? (considering variations between number of tiles per patient)
Thank you for your inquiry. Each model (ER, PR, ERBB2 times 5 folds) took approximately 9.5 days to train, using around 8,500 slides in total. The inference time per slide averages 9 seconds. We have included these details in the Methods section (line 669 in the revised manuscript).

6.5. Table 1: it is unclear why the performances are only presented on a per slide level and not per patient as well.

We understand your concern regarding the level of performance reporting. Although clinical decisions are informed at the patient level, there are several reasons why we reported results per slide rather than per patient:

1. **Multiple Labels per Patient:** Some patients had multiple IHC results showing distinct receptor statuses due to intratumor heterogeneity, multiple lesions with different molecular profiles, or multiple tumor excisions at different times. For instance, a core needle biopsy (CNB) and a subsequent surgical excision might show different receptor statuses (as seen in Fig 3a).

2. **Slide-Level Prediction:** The deep learning system predicts receptor expression based on the tumor region visible in the H&E slide. This localized prediction does not necessarily reflect patient-level expression, which could be influenced by other regions of the tumor or other tumors.
3. **Clinical Application:** The system is designed to be applied per H&E slide, where each result represents the slide itself. For example, in clinical practice, applying the system immediately to a biopsy is preferable to waiting for additional surgical specimens for aggregation. Likewise, for patients with multiple tumors, pathologists benefit from seeing predictions per tumor to inform decisions. If the system is designed to be applied per slide, the analysis should report the performance of the desired application per slide.
4. **Inconsistency Across Datasets:** Inconsistent labeling across datasets further complicates patient-level analysis. For example, TCGA often provides a single IHC result, while datasets like Carmel offer multiple IHC diagnoses in many cases.

Please note that for accurate statistical analysis, confidence intervals and P values were calculated assuming independence at the patient level. For example, bootstrapping was done at the patient level to maintain statistical rigor.

6.6. Line 137: are performances per slide or per patient here? Also, what are the CIs?
Thank you for your comment. Performance was indicated per slide. We revised the sentence and added the CIs (line 143 in the revised manuscript).

6.7 Ext Table 1: please clarify what is low and high positive for ER status?
We appreciate your request for clarification. The definitions of low and high positive ER status are provided in our introduction, based on current guidelines. Specifically, low positive ER status is defined as having an IHC score between 1% and 10% (inclusive), while high positive ER status is defined as an IHC score greater than 10%. In response to your comment, we have reiterated these definitions in the caption of Extended Data Table 1 for clarity.

6.8. Fig 5d: there is no color legend associated with the orange color
Thank you for your feedback. We added the color legend to the figure.

To all reviewers

Please find below a point-by-point response to all of your comments referring to the relevant changes made in the manuscript. We also attach a copy of the revised manuscript with all modifications in color for your convenience.

Reviewer #1 (Remarks to the Authors, Answers)

“3.2. Lines 702-713: I don't see the code on github for that part.
Thank you for pointing this out. We did not include the code for the proposed heatmap method on GitHub, as we did not consider it a novel contribution of our study. As mentioned in line 713 of the original manuscript, methods like GradCam yielded similar outcomes. However, if the reviewer feels it is necessary, we are happy to provide the code upon acceptance of the manuscript.”

--> Yes, it would be best to have everything in the github repository to facilitate reproduction by the users of the full study.

Thank you for your suggestion. We uploaded the heatmap code to the Github repository.

“Train and test": as such, it seems like we could fear data leaks for the study related to these datasets, right? We appreciate your concern regarding potential data leaks. Carmel-Test-Rescanned consists of slides from the Carmel-Test dataset that were rescanned using a different scanner. However, your concern regarding Carmel-Intratumor is valid. Carmel-Intratumor was designed to include patients from the entire Carmel dataset, not just Carmel-Test, to broaden the analysis. Here are some clarifications: (1) While patients from Carmel-Train are included in Carmel-intratumor, the system was applied to H&E slides from different blocks than those used in Carmel-Train. (2) Carmel-Train consists of five-folds, and we ensured that each model was applied to patients it was not trained on. While there is a potential risk of overfitting during hyperparameter tuning, this tuning was conducted solely using the TCGA training folds, not Carmel data. (3) The risk of overfitting in this context does not lead to an advantage. The system was applied to the Carmel-Intratumor dataset to identify tumor blocks predicted as ERpositive and PR-positive, even when the ground truth was negative. Overfitting to the ground truth status would actually result in identifying fewer potential false negative cases, which is contrary to our objectives.”

--> Were this clarified in the manuscript as well? If not please add it.

Thank you for your comment. We added the clarification to the manuscript (lines 624 - 630)

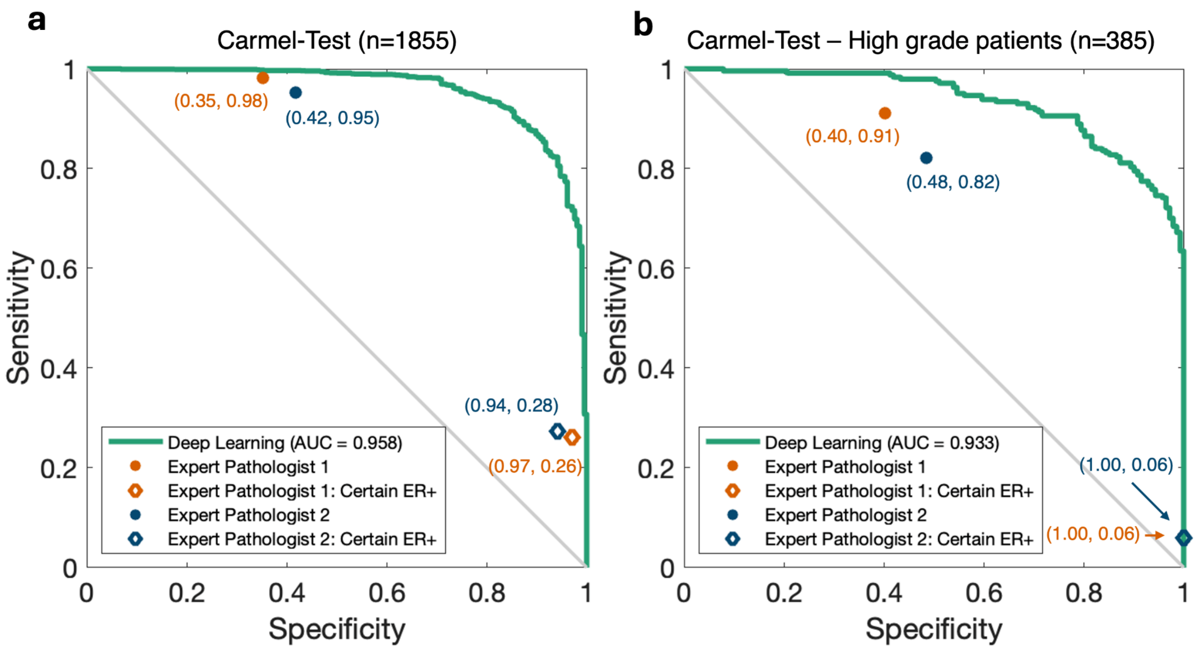
here are my thoughts re the responses to reviewer 2:

**** Answer to Question 2:
"I would also encourage the authors to perform a blinded study..."
- First, the AI performance is compared to pathologists' performances looking visually at "tile" images. This should also be done on "slide" images, since pathologist usually do these predictions looking at the whole slide, not single tile. It makes the comparison unfair.

- Second, the authors say "Although this comparison provides interesting insights, it is not the main focus of our study, and we did not add it to the paper."

>>>> Since reviewer 2 said Things like that would definitely increase the novelty of the study", I would suggest to add it to the manuscript (providing they add a comparison with estimation from WSI in addition to tiles)

Thank you for your suggestion. Following your comment, we asked two of our expert pathologists to guess the ER status of 250 slides from Carmel-Test. Their ability to predict the ER status was indeed much higher using WSI than using tile images. However, note that the models' performance results were also much higher when applied to WSI (in the previous revision we applied them to the same tile images). It is interesting to show that when constraining the analysis to high-grade patients the gap between the pathologists and the models was higher, probably because the pathologists heavily relied on the grade for their predictions. We added this experiment to the paper (lines 438-448)



**** Question 4:
"From a clinical perspective, the three biomarkers are typically used in combination to categorize patients ...."

>>>> Question is addressed in the rebuttal latter but the manuscript does not seem to have been altered accordingly.

Thank you for your feedback. We altered the text accordingly in lines 542 and 546-551.

*** Question 5:
"-The basis of the predictions—whether they pe.... "

>>>> same as before - the authors list 5 points to justify they choice - this should be added more clearly as such in the manuscript's method section as well

Thank you for your comment. We added the justifications for slide-level predictions to the manuscript in lines 754-764.